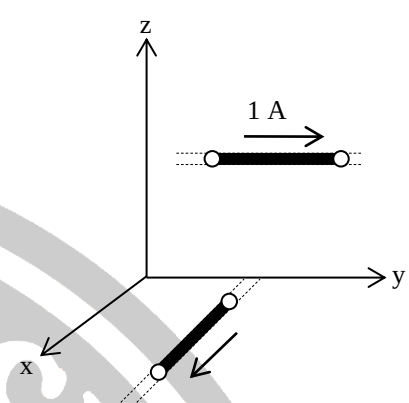




## Multiple Choice:

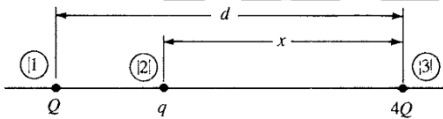
- Four 10nC positive charges are located in the  $z = 0$  plane at the corners of a square 8cm on a side. A fifth 10nC positive charge is located at a point 8cm distant from the other charges. Calculate the magnitude of the total force on this fifth charge for  $\epsilon = \epsilon_0$ .
  - $3.0 \times 10^{-4}$  N
  - $1.0 \times 10^{-4}$  N
  - $4.0 \times 10^{-4}$  N
  - $2.0 \times 10^{-4}$  N
- Point charges of 50nC each are located at A(1, 0, 0), B(-1, 0, 0), C(0, 1, 0), and D(0, -1, 0) in free space. Find the total force on the charge at A.
  - $20.5a_x \mu\text{N}$
  - $21.5a_x \mu\text{N}$
  - $22.5a_x \mu\text{N}$
  - $22.5a_x \mu\text{N}$
- Electrons are in random motion in a fixed region in space. During any  $1\mu\text{s}$  interval, the probability of finding an electron in a subregion of volume  $10^{-15} \text{ m}^3$  is 0.27. What volume charge density in  $\mu\text{C}/\text{m}^3$ , appropriate for such time durations, should be assigned to that subregion?
  - 42.2
  - 43.3
  - 44.4
  - 45.5
- Consider a pair of point charges in free space. Charge  $q = -300 \mu\text{C}$  is located at (2, 4, 5) m and charge  $Q = 10 \mu\text{C}$  is located at (1, 1, 3) m. What is the force (magnitude and direction) on  $q$ ?
  - 1.93 N directed from  $Q$  to  $q$
  - 1.93 N directed from  $q$  to  $Q$
  - 2.13 N directed from  $Q$  to  $q$
  - 2.13 N directed from  $q$  to  $Q$
- An electric field is given by  $E = xy^2 a_x - xz^2 a_y + xyz a_z$ . Calculate the  $x$  component of its curl at (2, 1, 3) in rectangular coordinates.
  - 15
  - 16
  - 17
  - 18
- $\nabla \times \nabla \times P$ , where  $P$  is a vector, is equal to \_\_\_\_\_.
  - $P \times \nabla \times P - \nabla^2 P$
  - $\nabla^2 P \times \nabla(\nabla \cdot P)$
  - $\nabla^2 P + \nabla \times P$
  - $\nabla(\nabla \cdot P) - \nabla^2 P$
- For static electric and magnetic fields in an in homogenous source-free medium, which of the following represents the correct form of two of Maxwell's equations?
  - $\nabla \cdot E = 0$  and  $\nabla \times B = 0$
  - $\nabla \cdot E = 0$  and  $\nabla \cdot B = 0$
  - $\nabla \times E = 0$  and  $\nabla \times B = 0$
  - $\nabla \times E = 0$  and  $\nabla \cdot B = 0$
- Vector potential is a vector:
  - whose curl is equal to the magnetic flux density
  - whose curl is equal to the electric field intensity
  - whose divergence is equal to the electric potential
  - which is equal to the vector product  $E \times H$
- The divergence of the vector field  $\vec{A} = x\hat{a}_x + y\hat{a}_y + z\hat{a}_z$  is:
  - 0
  - $1/3$
  - 1
  - 3
- Two infinitely long wires carrying current are as shown in the Figure below. One wire is in the  $y$ - $z$  plane and parallel to the  $y$ -axis. The other wire is in the  $x$ - $y$  plane and parallel to the  $x$ -axis. Which components of the resulting magnetic field are non-zero at the origin?
 
  - $x, y, z$  components
  - $x, y$  components
  - $y, z$  components
  - $x, z$  components
- The electric field component of a time harmonic plane EM wave traveling in a non-magnetic lossless dielectric medium has amplitude of 1 V/m. If the relative permittivity of the medium is 4, the magnitude of the time-average power density vector (in  $\text{W}/\text{m}^2$ ) is:
  - $1/30\pi$
  - $1/60\pi$
  - $1/120\pi$
  - $1/240\pi$
- The electric field strength at distant point,  $P$ , due to a point charge,  $+q$ , located at the origin, is  $100 \mu\text{V}/\text{m}$ . If the point charge is now enclosed by a perfectly conducting metal sheet sphere whose center is at the origin, then the electric field strength at the point,  $P$ , outside the sphere, becomes:
  - Zero
  - $100 \mu\text{V}/\text{m}$
  - $-100 \mu\text{V}/\text{m}$
  - $50 \mu\text{V}/\text{m}$
- Express Gauss' law in point form.
  - $\rho$
  - 1
  - $1/\rho$
  - $-\rho$



14. A metal sphere with 1 m radius and surface charge density of 10 coulombs/m<sup>2</sup> is enclosed in a cube of 10 m side. The total outward electric displacement normal to the surface of the cube is:

A.  $40\pi$  coulombs  
B.  $10\pi$  coulombs  
C.  $5\pi$  coulombs  
D. None of the above

15. Two positive charges Q and 4Q are separated by a distance d, in air. A third charge is so placed that the entire system is in equilibrium. Determine the magnitude and location of the third charge.



A.  $q = (-4/9)Q$ ,  $x = (2/3)d$   
B.  $q = (-4/9)Q$ ,  $x = (3/4)d$   
C.  $q = (-5/9)Q$ ,  $x = (2/3)d$   
D.  $q = (-5/9)Q$ ,  $x = (3/4)d$

16. According to the definition of dielectric strength of dielectric medium, which value of electric field give rise to an occurrence of dielectric breakdown phenomenon?

A. Maximum  
B. Minimum  
C. Initial  
D. Final

17. An electric field is given by  $E = xy^2 a_x - xz^2 a_y + xyz a_z$ . Evaluate the divergence of the E-field at (1, 2, 3).

A. 5  
B. 6  
C. 7  
D. 8

18. Find the magnitude of the electrostatic force between a 100  $\mu\text{C}$  charge at (-1, 1, -3) m and a 20  $\mu\text{C}$  at (3, 1, 0) m.

A. 0.63 N  
B. 0.72 N  
C. 0.85 N

D. 0.91 N

19. A positive charge and a negative charge are respectively located at (0, 1, 3) m and (3, 0, 0) m. Determine the direction cosines of the force on the positive charge.

A.  $3/\sqrt{19}$ ,  $1/\sqrt{19}$  and  $-3/\sqrt{19}$   
B.  $3/\sqrt{19}$ ,  $-1/\sqrt{19}$  and  $3/\sqrt{19}$   
C.  $-3/\sqrt{19}$ ,  $-1/\sqrt{19}$  and  $-3/\sqrt{19}$   
D.  $3/\sqrt{19}$ ,  $-1/\sqrt{19}$  and  $-3/\sqrt{19}$

20. A 100  $\mu\text{C}$  point charge ( $Q_1$ ), is located in a rectangular coordinates system at (1, 1, 1) m and another point charge ( $Q_2$ ) of 50  $\mu\text{C}$ , is at (-1, 0, -2) m. Find the vector force on the first charge.

A.  $1.718a_x + 0.859a_y + 2.577a_z$   
B.  $0.859a_x + 1.718a_y + 2.577a_z$   
C.  $2.577a_x + 0.859a_y + 1.718a_z$   
D.  $1.718a_x + 2.577a_y + 0.859a_z$

21. If a negative charge is absent corresponding to Faraday's experiment of concentric spheres, then where do the flux lines show termination?

A. At zero  
B. At unity  
C. At infinity  
D. At radial field

22. A sphere of 200-mm radius contains electrical charge of density  $2/(r \sin \theta)$  (C/m<sup>3</sup>). What is the total charge contained within the sphere?

A. 0.549 C  
B. 0.622 C  
C. 0.789 C  
D. 0.812 C

23. Given  $\rho_l(x, y, z) = 2x + 3y - 4z$  (C/m). Find the charge on the line segment extending from (2, 1, 5) to (4, 3, 6).

A. -10 C  
B. -20 C

C. -30 C  
D. -40 C

24. According to coulomb's law, what do the force between two point charges proportional to?

A. Addition of charges  
B. Difference of charges  
C. Product of charges  
D. Integration of charges

25. A circular disk of radius R has a surface charge density that increases linearly away from the center. The constant of proportionality being k. Determine the total charge on the disk.

A.  $(1/3)2\pi kR^3$   
B.  $(1/4)2\pi kR^3$   
C.  $(1/5)2\pi kR^3$   
D.  $\pi kR^3$

END